

第八周作业

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1. 习题四:

17.

$$\text{由题意得 } f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{others} \end{cases}$$

$$E(X) = \int_{-\infty}^{+\infty} xf(x)dx = \int_0^1 xdx = \frac{1}{2}$$

$$E(Y) = \int_{-\infty}^{+\infty} yf(y)dy = \int_5^{+\infty} ye^{-(y-5)}dy = 6$$

$\therefore X, Y$ 相互独立

$$\therefore E(XY) = E(X)E(Y) = 3$$

$$E(X^2) = \int_{-\infty}^{+\infty} x^2f(x)dx = \int_0^1 x^2dx = \frac{1}{3}$$

$$E(Y^2) = \int_{-\infty}^{+\infty} y^2f(y)dy = \int_5^{+\infty} y^2f(y)dy = 37$$

$$\therefore D(XY) = E(X^2Y^2) - E^2(XY) = E(X^2)E(Y^2) - (E(X)E(Y))^2 = \frac{10}{3}$$

$$D(2X - Y) = 4D(X) + D(Y) = 4(E(X^2) - E^2(X)) + (E(Y^2) - E^2(Y)) = \frac{4}{3}$$

19.

$$\therefore f_X(x) = \frac{1}{2\sqrt{\pi}}e^{-\frac{x^2+2x-1}{4}} = \frac{1}{\sqrt{2\pi} \cdot \sqrt{2}}e^{-\frac{(x-1)^2}{2 \cdot (\sqrt{2})^2}}f_Y(y) = \frac{1}{\sqrt{2\pi}}e^{-(0.5y^2+2y+2)} = \frac{1}{\sqrt{2\pi}}e^{-\frac{(y+2)^2}{2}}$$

$$\therefore X \sim N(1, 2), Y \sim N(-2, 1)$$

$$\therefore E(Z) = 2E(X) - E(Y) + 8 = 2 \times 1 - (-2) + 8 = 12$$

$$D(Z) = 4D(X) + D(Y) = 4 \times 2 + 1 = 9$$

20.

以线段左侧为原点, X, Y 分别为所取两点的坐标, $Y > X$, 则 $X \sim U(0, l), Y \sim U(0, l)$

$$f(x, y) = \begin{cases} \frac{1}{l^2}, & 0 < x < l, 0 < y < l \\ 0, & \text{others} \end{cases}$$

$$\begin{aligned}
2 \quad E(|X - Y|) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} |x - y| f(x, y) dx dy = \frac{1}{l^2} \int_0^l \int_0^l |x - y| dx dy = \frac{2}{l^2} \int_0^l \int_0^x (x - y) dy dx = \frac{l}{3} \\
E(|X - Y|^2) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} |x - y|^2 f(x, y) dx dy = \frac{1}{l^2} \int_0^l \int_0^l (x - y)^2 dy dx = \frac{l^2}{6} \\
\therefore D(|X - Y|) &= E(|X - Y|^2) - E^2(|X - Y|) = \frac{l^2}{18}
\end{aligned}$$

22.

$$(1) E(X) = \int_{-\infty}^{+\infty} x f(x) dx = \int_a^b x f(x) dx$$

$$\therefore \int_a^b x f(x) dx \geq \int_a^b a f(x) dx = a, \int_a^b x f(x) dx \leq \int_a^b b f(x) dx = b$$

$$\therefore a \leq E(X) \leq b$$

$$(2) \forall c \in R, D(X) \leq E((X - c)^2)$$

令 $g(c) = E((X - c)^2) = c^2 - 2cE(X) + E(X)$, 当 $c = E(X)$ 时 $g(c)$ 最小, 令 $c = \frac{a+b}{2}$

$$D(X) \leq E(X - (\frac{a+b}{2})^2) = \int_a^b (x - \frac{a+b}{2})^2 f(x) dx \leq \int_a^b (b - \frac{a+b}{2})^2 f(x) dx = \frac{(b-a)^2}{2}$$

2. 4.2-6 补充题:

由题意得

$$f_Z(z) = (1-p)f_X(z) + pf_{X+Y}(z) = (1-p)\frac{1}{\sqrt{2\pi}\sigma_1}e^{-\frac{(z-\mu_1)^2}{2\sigma_1^2}} + p\frac{1}{\sqrt{2\pi}\sqrt{\sigma_1^2+\sigma_2^2}}e^{-\frac{(z-(\mu_1+\mu_2))^2}{2(\sigma_1^2+\sigma_2^2)}}$$

$$E(Z) = \int_{-\infty}^{+\infty} z f(z) dz = \mu_1(1-p) + (\mu_1 + \mu_2)p$$

$$E(Z^2) = \int_{-\infty}^{+\infty} z^2 f(z) dz = (1-p)(\sigma_1^2 + \mu_1^2) + p(\sigma_1^2 + \sigma_2^2 + (\mu_1 + \mu_2)^2)$$

$$\therefore D(Z) = E(Z^2) - E^2(Z) = (1-p)\sigma_1^2 + p(\sigma_1^2 + \sigma_2^2) + p(1-p)\mu_2^2$$